

Average Value

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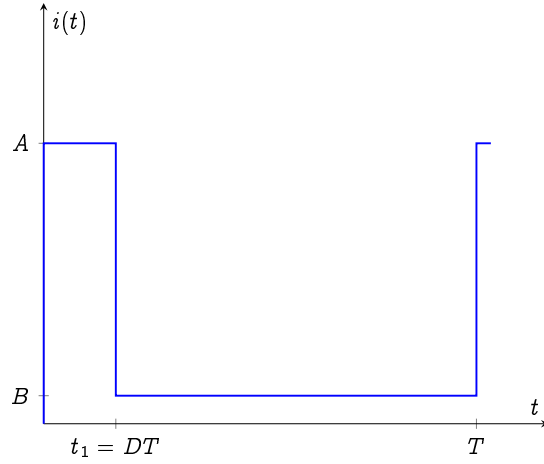


Figura 1:

$$i(t) = \begin{cases} A, & 0 \leq t \leq t_1 \\ B, & t_1 \leq t \leq T \end{cases} \quad (1)$$

The average value is given by:

$$I_{avg} = \frac{1}{T} \int_0^T i(t) dt \quad (2)$$

substituting (1) into equation (2) we get:

$$\begin{aligned} I_{avg} &= \frac{1}{T} \left[\int_0^{t_1} A dt + \int_{t_1}^T B dt \right] = \frac{1}{T} \left[At \Big|_0^{t_1} + Bt \Big|_{t_1}^T \right] \\ &= \frac{1}{T} [At_1 + B(T - t_1)] \end{aligned} \quad (3)$$

If we assume that $t_1 = DT$ where $D = \frac{t_1}{T}$ then:

$$I_{avg} = \frac{1}{T} [ADT + B(T - DT)] \quad (4)$$

$$I_{avg} = AD + B(1 - D) \quad (5)$$

If $A = 20 \text{ mA}$, $B = 60 \text{ } \mu\text{A}$, $t_1 = 20 \text{ } \mu\text{s}$, $T = 10 \text{ ms}$, then

$$D = \frac{t_1}{T} = \frac{20 \text{ } \mu\text{s}}{10 \text{ ms}} = 2 \times 10^{-3} \quad (6)$$

$$I_{avg} = (20 \text{ mA})(2 \times 10^{-3}) + (60 \text{ } \mu\text{A})(1 - 2 \times 10^{-3}) = 99.88 \text{ } \mu\text{A} \quad (7)$$

$$T \geq \frac{t_1(A + 0.01B)}{0.01B} = t_1 \left(100 \frac{A}{B} + 1 \right) \quad (8)$$

$$T \geq 666.69 \text{ ms} \quad (9)$$

$$I_{avg} \approx B(1 - D) \quad (10)$$

$$\approx B \left(1 - \frac{t_1}{T} \right) \quad (11)$$

$$V_o = \frac{10 \text{ mV}}{^{\circ}\text{C}} \times T[^{\circ}\text{C}] + 500 \text{ mV} \quad (12)$$

$$T[^{\circ}\text{C}] = \frac{V_o - 500 \text{ mV}}{\frac{10 \text{ mV}}{^{\circ}\text{C}}} \quad (13)$$

$$T[^{\circ}\text{C}] = 21.92 ^{\circ}\text{C} \quad (14)$$

$$\begin{array}{r}
 1 + x + 2x^2 + 3x^3 + 5x^4 + \dots \\
 1 - x - x^2 \overline{) 1} \\
 \underline{1 - x - x^2} \\
 x + x^2 \\
 \underline{x - x^2 - x^3} \\
 2x^2 - x^3 \\
 \underline{2x^2 - 2x^3 - 2x^4} \\
 3x^3 + 2x^4 \\
 \underline{3x^3 - 3x^4 - 3x^5} \\
 5x^4 + 3x^5 \\
 \vdots
 \end{array}$$

$$\begin{aligned}
D_I &= T - 0.55(1 - 0.01RH)(T - 14.50) \\
D_I - T &= -0.55(1 - 0.01RH)(T - 14.50) \\
T - D_I &= 0.55(1 - 0.01RH)(T - 14.50) \\
\frac{T - D_I}{T - 14.50} &= 0.55(1 - 0.01RH) \\
1 + \frac{14.50 - D_I}{T - 14.50} &= 0.55(1 - 0.01RH) \\
\frac{14.50 - D_I}{T - 14.50} &= 0.55(1 - 0.01RH) - 1 \\
[0.55(1 - 0.01RH) - 1][T - 14.50] &= 14.50 - D_I \\
T - 14.50 &= \frac{14.50 - D_I}{0.55(1 - 0.01RH) - 1}
\end{aligned}$$

$$T = 14.50 + \frac{14.50 - D_I}{0.55(1 - 0.01RH) - 1}$$

Where:

$$\begin{aligned}
&\frac{1}{T - 14.50} \left(\frac{T - D_I}{-T + 14.50} \right) \\
&\frac{-T + 14.50}{14.50 - D_I}
\end{aligned}$$

So

$$\frac{T - D_I}{T - 14.50} = 1 + \frac{14.50 - D_I}{T - 14.50}$$

$$D_I = T - 0.55 \cdot (1 - 0.01 \cdot RH)(T - 14.50) \quad (15)$$

Assuming that:

$$T = 38^\circ\text{C} \quad (16)$$

$$RH = 78\% \quad (17)$$

$$D_I = T - 0.55 \cdot (1 - 0.01 \cdot RH)(T - 14.50) \quad (18)$$

$$= 38^\circ\text{C} - (0.55)(1 - 0.01 \times 78\%)(38^\circ\text{C} - 14.50) \quad (19)$$

$$= 35.1565 \quad (20)$$

$$T = 14.50 + \frac{14.50 - D_I}{0.55(1 - 0.01RH) - 1} \quad (21)$$

$$= 14.50 + \frac{14.50 - 35.1565}{0.55(1 - 0.01 \times 78\%) - 1} \quad (22)$$

$$= 14.50 + \frac{-20.6565}{0.121 - 1} \quad (23)$$

$$= 14.50 + 23.5 \quad (24)$$

$$= 38^\circ\text{C} \quad (25)$$

However, in your equation the temperature would be:

$$T = \frac{D_I - (0.55)(14.5)(1 - 0.01RH)}{1 - 0.01RH} \quad (26)$$

$$= \frac{35.1565 - (0.55)(14.50)(1 - 0.01 \times 78 \%) }{1 - 0.01 \times 78 \%} \quad (27)$$

$$= \frac{35.1565 - (7.975)(1 - 0.78)}{1 - 0.78} \quad (28)$$

$$= \frac{35.1565 - 1.7545}{0.22} \quad (29)$$

$$= 151.83^\circ\text{C} \quad (30)$$

$$T = 14.50 + \frac{14.50 - D_I}{0.55(1 - 0.01RH) - 1} \quad (31)$$

$$= 14.50 + \frac{14.50 - 23}{0.55(1 - 0.01 \times 75) - 1} \quad (32)$$

$$= 14.50 + \frac{8.5}{0.8625} \quad (33)$$

$$= 24.35^\circ\text{C} \quad (34)$$

$$V_{ADC_swing,current} = \pm \frac{\sqrt{2}I_{RMS_{max}}(R_3 + R_8)}{CT_{TURNS_RATIO}} \quad (35)$$

$$V_{ADC_swing,voltage} = \pm \sqrt{2}V_{RMS} \left(\frac{R_{11}}{R_{11} + R_4 + R_5 + R_6 + R_7} \right) \quad (36)$$